
AI Footprint Calculator:

An Interactive Dashboard to Estimate Environmental Externalities of Generative AI Queries

Project Proposal

COL8381: Special Topics in Artificial Intelligence: Responsible AI

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1. Introduction and Problem Statement

Generative AI systems—large language models (LLMs) and image generation tools—are rapidly being integrated into everyday workflows. These systems run on large-scale computational infrastructure hosted in energy-intensive data centers. While prior research has examined the environmental footprint of ML training and large-scale deployment [1, 5], the environmental cost of *individual AI interactions* remains almost entirely invisible to end users.

Core Problem: Users currently have no accessible means to understand the energy consumption, carbon emissions, or water usage associated with their everyday AI interactions—despite these costs being real and cumulative at scale.

This project aims to build an interactive web-based dashboard that estimates and visualizes the approximate environmental footprint of AI queries. The tool will let users explore how usage patterns (frequency, model type, prompt length) affect environmental impact, promoting both awareness and more mindful AI usage.

2. Motivation

Responsible AI research has primarily focused on algorithmic bias, fairness, and explainability. Environmental sustainability, however, is an equally critical dimension of AI accountability [6, 7]. Although a single query has a small footprint, aggregate usage at scale leads to substantial cumulative impact on energy demand, carbon emissions, and water consumption.

Data-center electricity consumption is projected to grow significantly with the rapid expansion of AI-driven services [8]. Despite this, there are no user-facing tools that communicate environmental consequences of everyday AI use in an interpretable way. This project addresses that gap by increasing awareness of hidden environmental costs, encouraging mindful AI usage, and providing an educational interface for understanding Responsible AI trade-offs.

3. Related Work

Patterson et al. [1] analyzed carbon emissions of ML workloads, showing that inference at scale contributes significantly to environmental footprint. Luccioni et al. [2] provided empirical per-query measurements across model families and emphasized the need for standardized reporting. Dodge et al. [3] showed that hardware efficiency, model size, and deployment region can shift carbon intensity by up to 30×. Ren et al. [4] quantified freshwater consumption of data center cooling, estimating roughly 500 mL of water per 20–50 GPT-3-class responses. GPTFootprint [9] demonstrated feasibility of browser-level

environmental estimation for generative AI.

While prior work focuses on system-level measurement, there is limited exploration of *user-facing interactive tools* translating these impacts into per-query estimates. Table 1 positions this project against *closest prior work*.

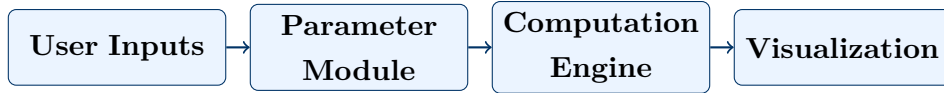
Table 1: Positioning relative to related work.

Feature	Luccioni et al.	Dodge et al.	GPTFootprint	This Work
Energy + Carbon	✓	✓	✓	✓
Water Usage	—	—	—	✓
User-Facing Tool	—	—	✓	✓
Usage Case Studies	—	—	—	✓
Sensitivity Analysis	—	—	—	✓
No Setup Required	—	—	✓	✓

4. Proposed Approach

4.1. System Design

The tool is a single-page web application requiring no installation. It consists of three modules: a **parameter module** encoding literature-derived coefficients, a **computation engine** performing real-time estimation, and a **visualization layer** rendering results interactively.



4.2. Inputs and Outputs

Inputs: number of queries, model type (small/large LLM or image generation), prompt length (token estimate), and usage frequency (daily/weekly/monthly).

Outputs: estimated energy consumption (kWh), carbon emissions (gCO₂e), and water usage (mL), alongside real-world equivalence metrics such as smartphone charging cycles, distance driven, and glasses of water.

4.3. Estimation Methodology

All estimates are derived from literature-calibrated parameters. Energy per query E_q (kWh) is modeled as:

$$E_q = \alpha_m \cdot \beta_\ell \cdot \gamma_h \cdot \frac{1}{\eta_{\text{PUE}}} \quad (1)$$

where α_m is the base energy coefficient for model class m [2], β_ℓ is a prompt-length scaling factor, γ_h is a hardware efficiency factor, and η_{PUE} is the Power Usage Effectiveness

of the data center (typical range 1.1–1.6). Carbon emissions are then $C_q = E_q \cdot I_{\text{grid}}$, where I_{grid} is the regional grid carbon intensity [3]. Water usage follows Ren et al. [4]: $W_q = E_q \cdot \kappa_{\text{WUE}} \cdot \phi_{\text{cool}}$.

Reference parameter ranges by model type are given in Table 2. All estimates are shown with uncertainty ranges (min/median/max) to reflect cross-study variability.

Table 2: Literature-derived parameter ranges by model type.

Model Type	Energy (kWh/query)	Carbon (gCO ₂ e/query)	Water (mL/query)
Small LLM (<7B)	0.0001–0.001	0.01–0.1	0.1–1.0
Medium LLM (7–70B)	0.001–0.005	0.1–0.5	1.0–5.0
Large LLM (>70B)	0.005–0.02	0.5–2.0	5.0–20.0
Image Generation	0.01–0.1	1.0–10.0	10–100

Sources: [2, 4, 3]

4.4. Visualization Features

- **Cumulative Projections:** Time-series charts showing daily, monthly, and annual cumulative impact to convey scale effects invisible at the per-query level.
- **Sensitivity Analysis:** Interactive sliders for PUE and prompt length, showing how these factors shift the final estimates.
- **Real-World Equivalences:** Abstract numbers translated into relatable comparisons (e.g., “equivalent to charging your phone X times” or “driving Y km”).

4.5. Usage Case Studies

To ground the tool in realistic contexts, we will evaluate two representative user profiles:

- **Student workflow:** A student using a medium LLM (e.g., ChatGPT) for roughly 30–50 queries/day for assignments, coding help, and summarization—estimated on a daily, monthly, and semester basis.
- **AI-heavy professional:** A developer or researcher running 300–500 queries/day across large LLMs and image generation tools for prototyping, debugging, and content creation.

These case studies will illustrate the contrast in footprint between casual and intensive AI usage, make scale effects tangible, and serve as a validation check on the estimation framework.

5. Limitations, Contributions, and Plan

5.1. Limitations

The estimates are necessarily approximate. Key sources of uncertainty include: (i) infrastructure heterogeneity—PUE and cooling technology vary across data centers and are not publicly disclosed; (ii) model opacity—inference costs depend on architecture details not publicly documented; (iii) cross-study inconsistency in measurement methodology; and (iv) rapid hardware improvement that may render parameters outdated. The tool will communicate these explicitly via confidence intervals and tooltips, and will frame itself as an *educational estimator*, not an auditing instrument. Training, fine-tuning, and embodied hardware carbon are out of scope.

5.2. Expected Contributions

- A publicly usable browser-based Responsible AI awareness tool covering energy, carbon, and water.
- A documented estimation framework translating system-level measurements into per-query estimates.
- Visualization of cumulative scale effects supporting individual and institutional awareness.

5.3. Project Plan

Table 3: Project milestones.

Phase	Milestone	Deliverables
1	Literature Survey & Parameter Calibration	Parameter table with uncertainty ranges
2	System Design	UI wireframes; interaction flow
3	Core Implementation	Estimation module; baseline visualizations
4	Advanced Features	Sensitivity analysis; real-world equivalences
5	Case Studies & Documentation	Student & professional workflow analysis; final report

References

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